

# 3D Shapes - Lesson Plan

Cube \* Cuboid \* Sphere \* Cone

Grade 1 | 30 Minutes | Beginner | Planrs Academy

|                        |                   |                      |                            |
|------------------------|-------------------|----------------------|----------------------------|
| Duration<br>30 Minutes | Level<br>Beginner | Category<br>Geometry | Grade<br>Grade 1 (Age 5-7) |
|------------------------|-------------------|----------------------|----------------------------|

## Learning Objectives

| LO   | Objective                                                                               |
|------|-----------------------------------------------------------------------------------------|
| LO-1 | Name and identify the four basic 3D shapes: <b>Cube, Cuboid, Sphere, and Cone</b> .     |
| LO-2 | Describe each shape using simple words: flat face, round, pointy top, no corners.       |
| LO-3 | Match each shape to at least one everyday <b>Indian object</b> found at home or school. |
| LO-4 | Sort a group of mixed objects into four shape categories <b>independently</b> .         |
| LO-5 | Distinguish 3D from 2D shapes: <b>"Can you hold it? Then it is 3D!"</b>                 |

## Key Concepts and Indian Context Examples

| Shape                 | Key Properties                                            | Indian Examples                                                        |
|-----------------------|-----------------------------------------------------------|------------------------------------------------------------------------|
| CUBE (dice emoji)     | 6 equal flat faces 8 corners   12 edges All sides equal   | Ludo dice, Sugar cube, Rubiks Cube, Wooden alphabet block, Paneer cube |
| CUBOID (box emoji)    | 6 faces (diff sizes) 8 corners   12 edges Sides NOT equal | Tiffin/Dabba, Book, Biscuit packet, Mithai box, Brick (eent)           |
| SPHERE (orange emoji) | No flat face No corners, no edges Rolls in all directions | Cricket ball, Laddoo, Orange/Amla, Marble (kancha), Cotton batti       |
| CONE (icecream emoji) | 1 circular flat face 1 curved surface 1 pointy tip (apex) | Kulfi cone, Birthday party hat, Rangoli cone, Vendor snack cone        |

Session Outline with Timing (30 Minutes)

| Time      | Segment                              | Activity / Teacher Notes                                                                                         | Type         |
|-----------|--------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------|
| 0-3 min   | <b>HOOK</b> - What is in the Bag?    | Show bag with lunchbox, ball, dice. Ask students to guess the shape. Rule: "Can you hold it? It is 3D!"          | Story Intro  |
| 3-6 min   | <b>2D vs 3D</b> - Flat or Fat?       | Circle (drawn) vs orange. Square vs dice. Use real objects if available. Introduce the "hold it" rule.           | Concept      |
| 6-10 min  | <b>CUBE</b> - Meet the Dice!         | Show real dice. Count faces (6). Rhyme: "Six faces all the same, Cube is its cool 3D name!" Show sugar cube.     | Teach + Show |
| 10-14 min | <b>CUBOID</b> - The Lunchbox Shape!  | Show tiffin. Compare with cube - same face count, different sizes. Show book, brick, biscuit pack.               | Teach + Show |
| 14-18 min | <b>SPHERE</b> - The Rolling Laddoo!  | Roll ball across table. "Any flat part? No!" Show cricket ball, orange, marble. Checkpoint: "Point to a sphere!" | Teach + Show |
| 18-22 min | <b>CONE</b> - Ice-cream Hat!         | Show kulfi cone / party hat. "Pointy top, round bottom." Make paper cone. Ask: "Does it roll in a circle?"       | Teach + Show |
| 22-26 min | <b>ACTIVITY</b> - Sort the Shapes!   | Digital/physical sort: 8 objects into 4 baskets. Laddoo, book, cone hat, dice, orange, biscuit box, ball.        | Interactive  |
| 26-28 min | <b>RECAP SONG</b> - Shapes Around Us | 2-min song with actions: roll arms (sphere), stack palms (cube/cuboid), point up (cone). Repeat chorus twice.    | Audio/Visual |
| 28-30 min | <b>BADGE</b> - Shape Star!           | Exit Q: "Tell me one sphere at home!" Award "3D Shape Explorer" badge. Celebrate all completions warmly.         | Celebration  |

Tone, Style and Language Guide

|                         |                                                                                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Voice and Praise</b> | Warm, playful, encouraging. "Waah! Sahi jawab!" / "Shabash!" Celebrate every win. Address students directly.            |
| <b>Language Level</b>   | Simple English, max 8-10 words per sentence. No passive voice. Repeat shape names 3+ times per segment.                 |
| <b>Visual Style</b>     | Bright saturated colours. Cartoon/illustrated aesthetic. Diverse Indian children. Friendly faces on shapes.             |
| <b>Music</b>            | Light Indian folk-fusion background. Dholak beats for transitions. Clear voice-over. No lyrics during concept delivery. |
| <b>Pacing</b>           | Change visual every 15 sec. New object every 20-30 sec. Built-in pause prompts: "Now YOU try - tap the sphere!"         |

## Things to Avoid

| Content                                                                                                                                                                                                                                                                                      | Design and Interaction                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Do NOT introduce Cylinder or Pyramid (Grade 2+)<br>No technical terms: vertex count, Euler formula<br>Never say "a sphere is a circle"<br>No culturally irrelevant objects (igloo, sombrero)<br>No red X marks or "WRONG" text for incorrect answers<br>No more than 4 shapes in this module | No white/plain backgrounds - illustrated only<br>No text-heavy slides - visuals carry the meaning<br>No font below 18pt on screen, 16pt on worksheets<br>No timed pressure activities for this age group<br>No activities requiring reading full sentences<br>No gender-biased role assignment in illustrations |

## Formative Assessment Checkpoints

| Checkpoint        | When                   | Method                         | What to Look For                                                             |
|-------------------|------------------------|--------------------------------|------------------------------------------------------------------------------|
| <b>CP-1 Entry</b> | Mins 0-3               | Verbal Q and A<br>Show objects | Can student tell flat vs solid? Do they know any 3D shape name?              |
| <b>CP-2 Mid</b>   | Min 18 After<br>Sphere | Point-to activity              | Can student point to cube, cuboid, sphere correctly from a group of objects? |
| <b>CP-3 Exit</b>  | Mins 28-30             | Voice question or emoji        | "Tell me one sphere at home." Student names at least 1 correct object.       |

## Differentiation Strategies

| For Struggling Students                                                                                                                                                                                                                                                 | For Advanced Students                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use real objects, not pictures only<br>Focus on just 2 shapes first: Cube and Sphere<br>Use body actions: roll arms = sphere, flat palms = cube<br>Allow pointing instead of naming verbally<br>Pair with buddy for sort activities<br>Repeat shape song multiple times | Count corners and edges of each shape<br>Ask: "Which shapes can roll? Which cannot?"<br>Challenge: find 2 more objects at home per shape<br>Introduce Cylinder as bonus "surprise shape"<br>Ask: "What shape is half a sphere?" (hemisphere)<br>Create a shape journal with drawings |

## Success Criteria

| OK  | Student Outcome or Engagement Metric              | Target       |
|-----|---------------------------------------------------|--------------|
| YES | Students name all 4 shapes without prompt         | 80%          |
| YES | Match at least 3 Indian objects to correct shapes | 80%          |
| YES | Explain one property per shape in own words       | 70%          |
| YES | Activity (digital/physical sort) completion rate  | >75%         |
| YES | Post-quiz score (5 picture Qs, star rating)       | 3 of 5 stars |
| YES | Session engagement (watch time or participation)  | >22 min      |
| YES | Student happiness emoji rating after session      | 4+ of 5      |

Home Activity - Parent Note

After this lesson, ask your child to walk around the house and find: **1 cube, 1 cuboid, 1 sphere** and **1 cone**. Let them explain why each object belongs to that shape. Takes only 5-10 minutes!

*Suggested objects: dice, lunchbox, orange, ice-cream cone or party hat.*

Pre-Delivery Checklist

|                                                                                                |
|------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> All 4 shapes covered with minimum 3 Indian object examples each       |
| <input type="checkbox"/> No copyrighted brand logos or images used in any visual               |
| <input type="checkbox"/> Voice-over reviewed for Hindi word pronunciation                      |
| <input type="checkbox"/> Digital activity tested on tablet and low-bandwidth (2G) connection   |
| <input type="checkbox"/> Worksheet printed and verified for black-and-white print clarity      |
| <input type="checkbox"/> Lesson plan reviewed by at least one practising Grade 1 teacher       |
| <input type="checkbox"/> Hindi subtitles and captions attached where applicable                |
| <input type="checkbox"/> Files named: [CourseID]_Grade1_3DShapes_[OutputType]_v1               |
| <input type="checkbox"/> Curriculum Lead sign-off obtained before LMS upload                   |
| <input type="checkbox"/> Accessibility confirmed: font size, contrast ratio, touch target size |

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